Spark Plugs - (850/950) Briggs 491055S
Spark Plugs - (600/650/700 EFI) Briggs 496018S
Spark Plugs - (800) Kawasaki 92070-7004
Spark Plugs - (850) ODG - RC 12YC
Spark Plugs - (650) ODG RC 12YC
Spark Plug Torques:

Kawaski 800 - 18.5 ft·lb (25 Nm)
Briggs 850/950 - 180 in.- lbs. (20 Nm)
Briggs 600/650/700 - 180 in. lbs. (20 Nm)
ODG 850/650 - 221in.- lbs.(25 Nm)
Remove and inspect the spark plugs after every 100 hours of operation. Clean the plugs and reset the gap as detailed in the engine owner's manual.

Replace the spark plugs if the electrodes are corroded or damaged or if the insulator is cracked. Use the correct plug for the engine as detailed in the engine owner's manual.

Engine Cooling & Exhaust System

Engine cooling air is drawn in on the right side of the engine compartment and expelled with the exhaust on the left side. Keep all ducting and screening in place and free from any debris or mud.

Coolant Recommendations - Kawasaki 800

Use equal parts of ethylene glycol (anti-freeze) and water only. Distilled or deionized water is recommended, especially in areas where the water contains a high mineral content. Propylene glycol based anti-freeze is ***not*** recommended.

This mixture will provide protection from -37º C (-34º F) to 108º C (226º F). For protection and use outside the indicated temperature limits, follow the anti-freeze manufacturers instructions on the container, but do not exceed 70% anti-freeze.

DO NOT use anti-freeze with stop-leak additive(s), or put any other additives in the cooling system.

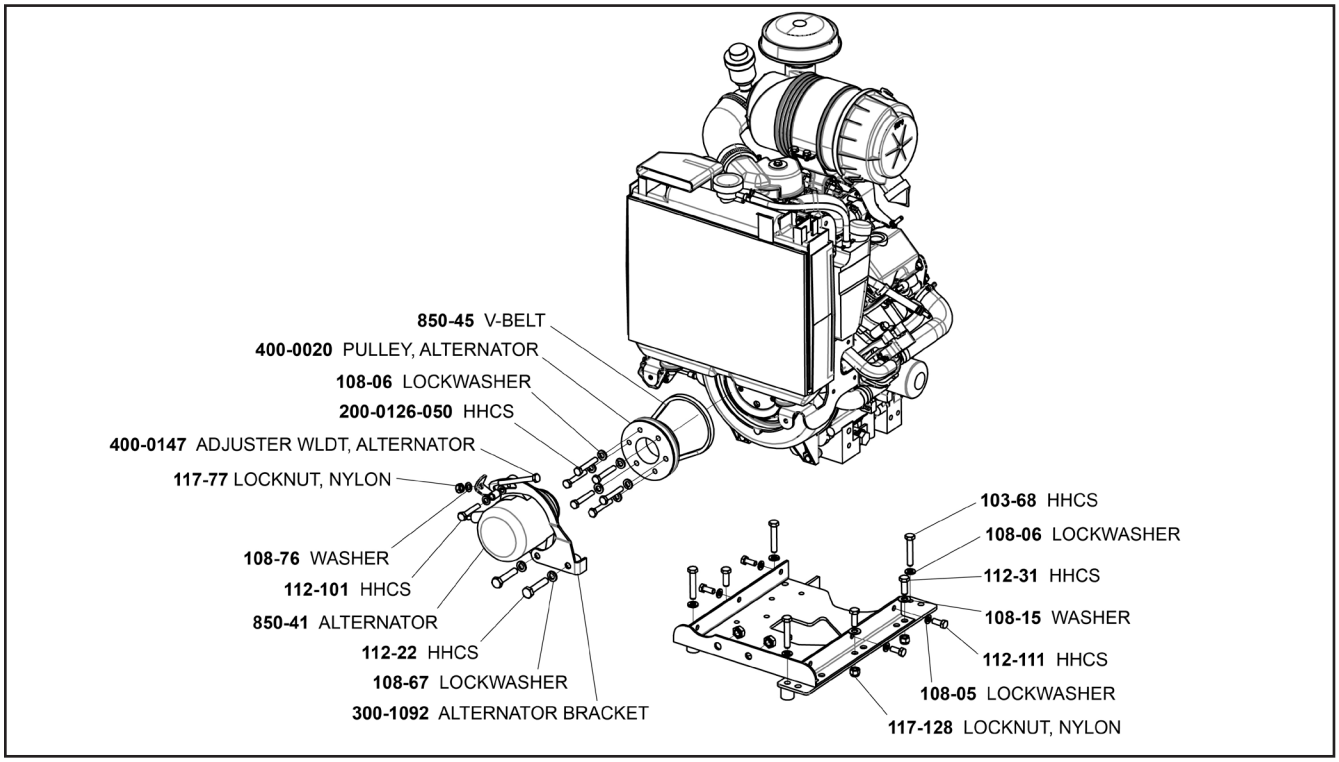
Type: Permanent type of anti-freeze; green coloured
Mixed Ratio: 50% mixed
Freezing Point: -35º C (-31º F)

Coolant Capacity: FD851D - 2.7L (2.9 US.qt.)

IMPORTANT

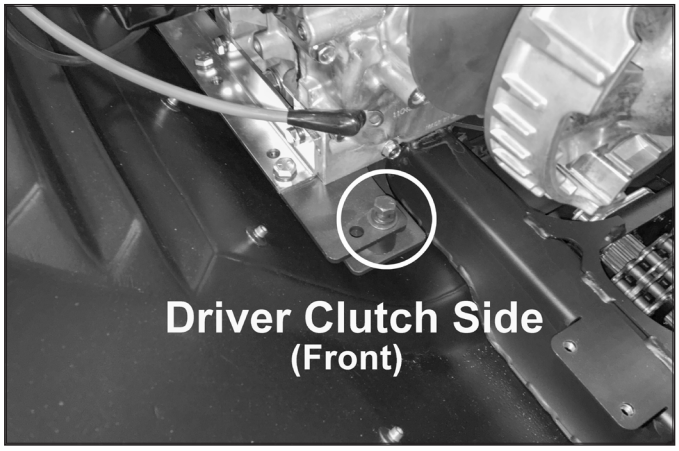
Refer to the diagram for detailed information on the engine mounting hardware.

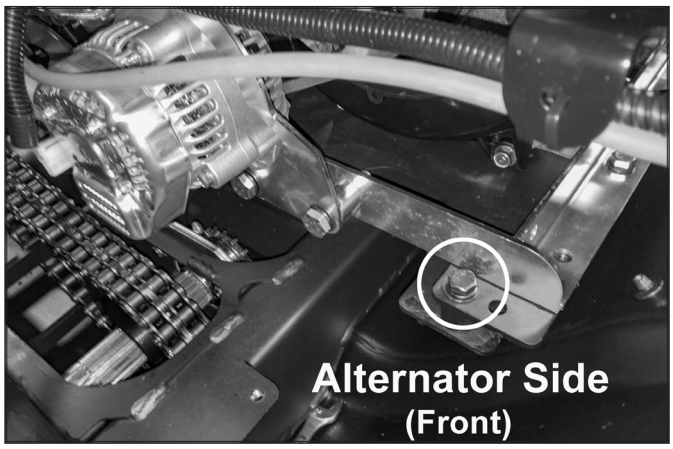
Removing The Kawasaki 800 Engine From The Vehicle

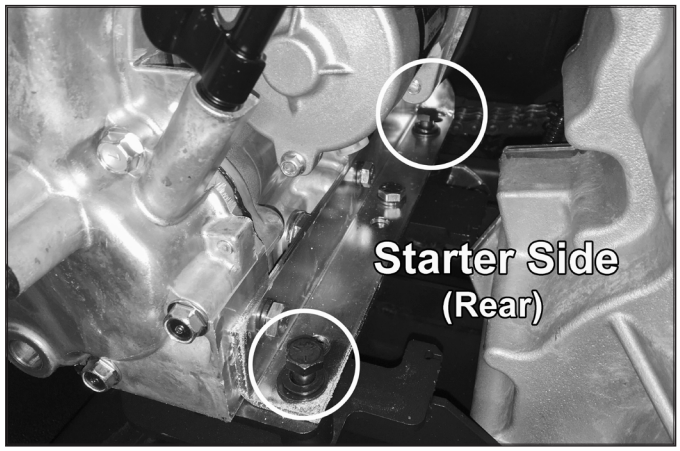


1. Disconnect the battery positive (+) and negative (-) cables from the battery.
2. Remove the heat deflector shield covering the muffler.
3. Locate the tailpipe assembly and detach the connecting springs between the tailpipe assembly and muffler. Pull the tail pipe assembly out through the hood area.
4. Disconnect the fuel line and return fuel line from the engine.
5. Disconnect the ground cable at the engine.
6. Disconnect the throttle cable at the engine cable retainer.
7. Disconnect the intake hose from the air filter cannister.
8. Disconnect fuel vent from fitting on engine.
9. Disconnect engine connector plugs in the main wire harness located at the front of the engine.
10. Disconnect all engine sensor wiring between engine and main vehicle wire harness.
11. Disconnect the wiring at the starter motor.

12. There are only four (4) bolts & lock washers securing the engine to the main frame front & rear. **Take note of which holes in the engine mounting frame are used for installation later.**







13. Remove all four (4) mounting bolts using a 9/16" socket and ratchet.

Lifting The Engine From The Vehicle

CAUTION

This procedure requires the use of an overhead hoist. Do NOT attempt to lift out by hand. Ensure all components attached to the engine have been detached. This may include any accessories that may be installed to the vehicle. Lift out slowly and carefully. If you wish to remove the muffler from the engine before raising it out, you may do so as an optional procedure. You may find it easier to maneuver the engine up and out with the muffler removed, but it is not necessary. Protect the upper body with heavy blankets to avoid any damage to the plastic!

WARNING

Ensure all hoist chains or belts are secure, and hooks are locked into place. Keep all fingers and hands clear of the area from which the engine is being lifted.

13. Attach the overhead hoist to the engine. There are lift hooks provided on each engine.
14. Slowly and gently raise the engine through the hood area with the overhead hoist. Watch for any wiring they may have been overlooked in disconnecting which may include some optional accessory kit wiring. Some manoeuvring is required as the engine does not necessarily come straight up and out. Avoid damage to any other components as well as the upper body.
15. Place the engine on a clean work bench for service required.
16. Remove the front and rear engine mount brackets from the engine block.





Installing The Kawasaki 800 Engine

1. Install the front and rear engine mount brackets to the engine block using four (4) HHCS M8x1,25x20 (8.8) ZN and four (4) LOCK WASHER 5/16 SPLIT-RING BNZ and torque to **225.4 IN-LBS (Dry, Non-Lubricated Condition).**
2. Slowly and gently lower the engine through the hood area. Some maneuvering is required as the engine does not necessarily drop straight in. Avoid damage to any vehicle components as well as the upper body.

⚠ IMPORTANT

Once the engine is seated into place on the lower frame, ensure that the rubber rad seal is properly sealing at the face of the radiator.

3. Align the engine mounted frame holes with the holes in the lower main frame and reinstall each of the four (4) HHCS 3/8-16x2.0 GR5 BNZ & 108-06 Lock washers. Secure and torque each fastener to **31 FT-LBS** (Dry, Non-Lubricated Condition)
4. Reconnect all engine sensor wiring between engine and main vehicle wire harness.
5. Reconnect engine connector plugs in the main wire harness located at the front of the engine.

6. Reconnect fuel vent from fitting on engine.
7. Reconnect the intake hose from the air filter cannister.
8. Reconnect the throttle cable at the engine cable retainer.
9. Reconnect the ground cable at the engine.
10. Reconnect wiring at the starter motor.
11. Reconnect the fuel line and return fuel line from the engine.
12. Locate the tailpipe assembly and reattach to the muffler with connecting springs.
13. Reinstall the heat deflector shield over the muffler.
14. Reconnect the battery positive (+) and negative (-) cables.

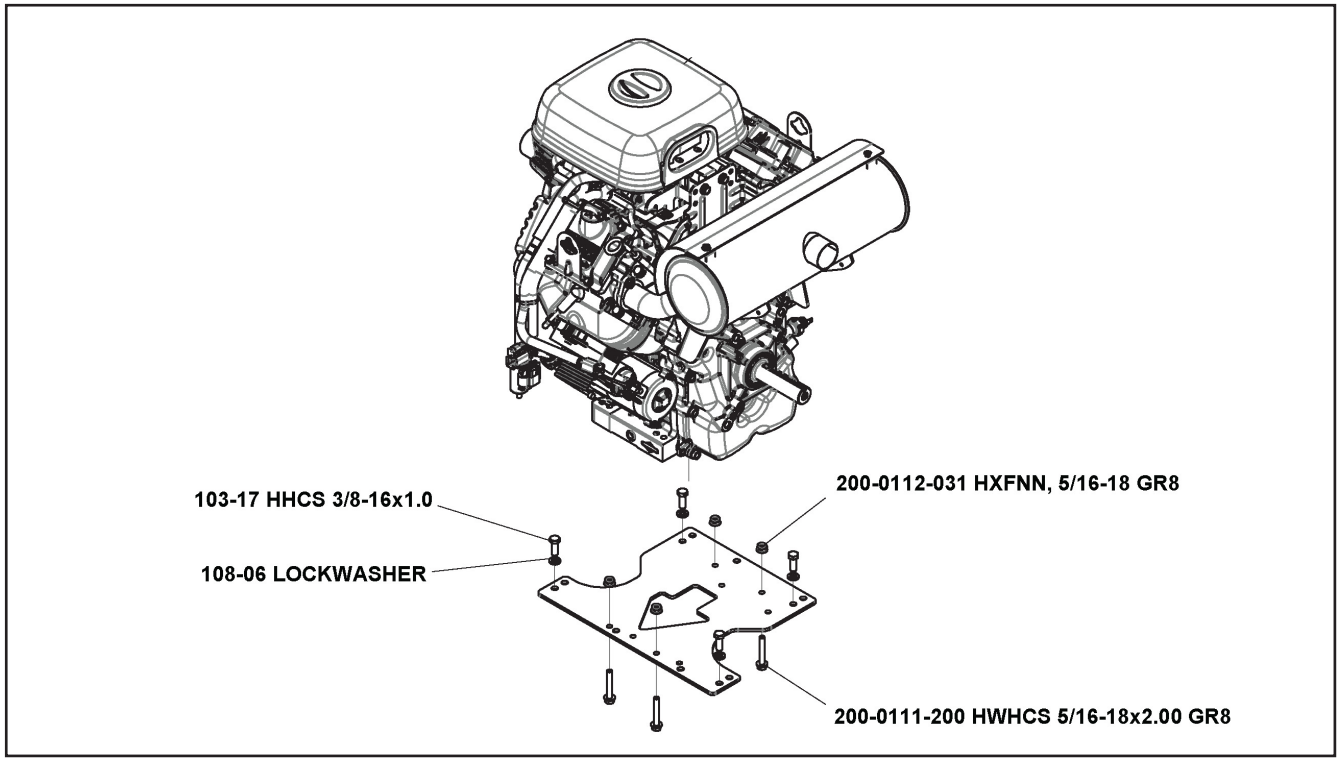
⚠ IMPORTANT

Reconnect the Positive (+) cable first, followed by the negative cable.

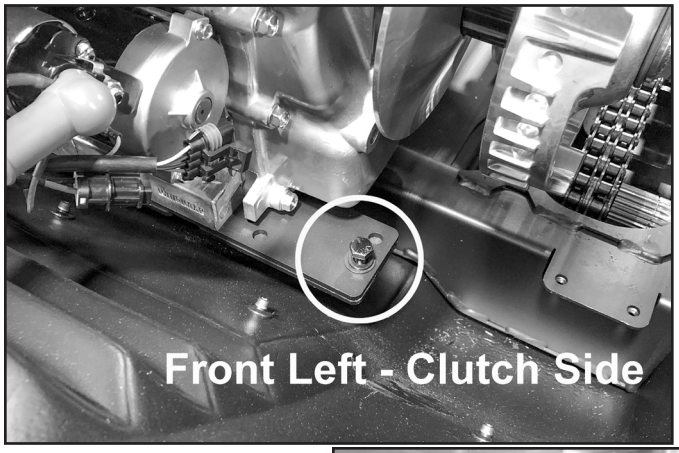
⚠ IMPORTANT

Refer to the diagram for detailed information on the engine mounting hardware.

Removing The Briggs Big Block 850/950 Engine From The Vehicle.



1. Disconnect the battery positive (+) and negative (-) cables from the battery.
2. Remove the heat deflector shield covering the muffler.
3. Locate the tailpipe assembly and detach the connecting springs between the tailpipe assembly and muffler. Pull the tail pipe assembly out through the hood area.
4. Disconnect the fuel line and return fuel line from the engine.
5. Disconnect the ground cable at the engine.
6. Disconnect the throttle cable at the engine cable retainer.
7. Disconnect the intake hose from the air filter cannister.
8. Disconnect fuel vent from fitting on engine.
9. Disconnect engine connector plugs in the main wire harness located at the front of the engine.
10. Disconnect the wiring at the starter motor.
11. Disconnect all engine sensor wiring between engine and main vehicle wire harness.
12. There are only four (4) bolts & lock washers securing the engine to the main frame front & rear. **Take note of which holes in the engine mounting frame are used for installation later.**
13. Remove all four (4) mounting bolts using a 9/16" socket and ratchet.







Lifting The Engine From The Vehicle

CAUTION

This procedure requires the use of an overhead hoist. Do NOT attempt to lift out by hand. Ensure all components attached to the engine have been detached. This may include any accessories that may be installed to the vehicle. Lift out slowly and carefully. If you wish to remove the muffler from the engine before raising it out, you may do so as an optional procedure. You may find it easier to maneuver the engine up and out with the muffler removed, but it is not necessary. Protect the upper body with heavy blankets to avoid any damage to the plastic!

WARNING

Ensure all hoist chains or belts are secure, and hooks are locked into place. Keep all fingers and hands clear of the area from which the engine is being lifted.

14. Attach the overhead hoist to the engine. There are lift hooks provided on each engine.
15. Slowly and gently raise the engine through the hood area with the overhead hoist. Watch for any wiring they may have been overlooked in disconnecting which may include some optional accessory kit wiring. Some manoeuvring is required as the engine does not necessarily come straight up and out. Avoid damage to any other components as well as the upper body.
16. Place the engine on a clean work bench for service required.
17. Remove the engine from the engine mount plate.





Installing The Briggs 850/950 Engine To The Vehicle

1. Install the engine to the mounting plate with four (4) HXFNN, 5/16-18 GR8 YZN and four (4) HWHCS 5/16-18x2.00 GR8 YZN. **Torque to 295 IN-LBS (Dry, Non-Lubricated Condition)**

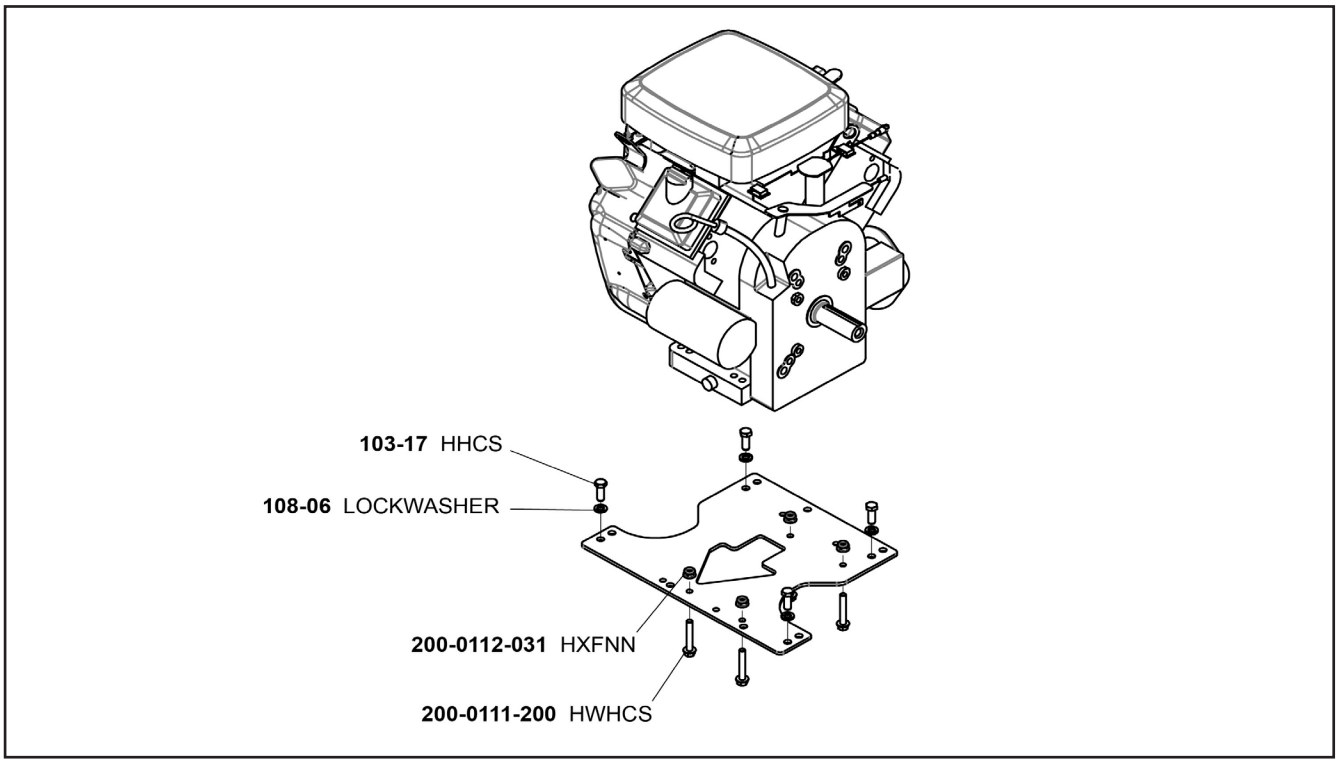
2. Slowly and gently lower the engine through the hood area. Some maneuvering is required as the engine does not necessarily drop straight in. Avoid damage to any vehicle components as well as the upper body.
3. Align the engine mounted frame holes with the holes in the lower main frame and reinstall each of the four (4) HHCS 3/8-16x2.0 GR5 BNZ & 108-06 Lock washers. Secure and torque each fastener to **31 FT-LBS** (Dry, Non-Lubricated Condition)
4. Reconnect all engine sensor wiring between engine and main vehicle wire harness.
5. Reconnect engine connector plugs in the main wire harness located at the front of the engine.
6. Reconnect fuel vent from fitting on engine.
7. Reconnect the intake hose from the air filter cannister.
8. Reconnect the throttle cable at the engine cable retainer.
9. Reconnect the ground cable at the engine.
10. Reconnect the fuel line and return fuel line from the engine.
11. Reconnect the wiring at the starter motor.
12. Locate the tailpipe assembly and reattach to the muffler with connecting springs.
13. Reinstall the heat deflector shield over the muffler.
14. Reconnect the battery positive (+) and negative (-) cables.

⚠ IMPORTANT

Reconnect the Positive (+) cable first, followed by the negative cable

Refer to the diagram for detailed information on the engine mounting hardware.

Removing The Briggs 650/700 EFI Briggs Engine From The Vehicle.



1. Disconnect the battery positive (+) and negative (-) cables from the battery.
2. Remove the heat deflector shield covering the muffler.
3. Locate the tailpipe assembly and detach the connecting springs between the tailpipe assembly and muffler. Pull the tail pipe assembly out through the hood area.
4. Disconnect the fuel line and return fuel line from the engine.
5. Disconnect the ground cable at the engine.
6. Disconnect the throttle cable at the engine cable retainer.
8. Disconnect fuel vent from fitting on engine.
9. Disconnect engine connector plugs in the main wire harness located at the front of the engine.
10. Disconnect all engine sensor wiring between engine and main vehicle wire harness (Briggs 700 models only)
11. Disconnect the wiring at the starter motor.
12. There are only four (4) bolts & lock washers securing the engine to the main frame front & rear. **Take note of which holes in the engine mounting frame are used for installation later.**